

ECE 434: DSP

Problem Set 1

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1. Determine the mod-4 and mod-4 reductions of the length 8 signal vector:

$$x = [1, 2, -2, 3, 4, -2, -1, 1]^T \quad (1)$$

Solution:

Mod-N reduction involves partitioning the vector into sequences of length N and then summing the sequences element-wise.

Thus, mod-4 reduction of x is given by:

$$x_{mod4} = [1 + 4, 2 - 2, -2 - 1, 3 + 1]^T = [5, 0, -3, 4]^T \quad (2)$$

Similarly, mod-3 reduction of x is given by:

$$x_{mod2} = [1 + 3 - 1, 2 + 4 + 1, -2 - 2]^T = [3, 7, -4]^T \quad (3)$$

Note that for the mod-3, the third sequence has only 2 elements, so we assume the third element is a 0 (pad zeroes where required).

2. Let $x = [1, 2, 2, 1, 2, 1, 1, 2]$, compute the 4-point DFT using the matrix form definition **Solution:**

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j & 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \\ 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (4)$$

Now, compute the mod-4 reduction of x and then compute the 4-point DFT of the reduced signal. Compare the results with the previous DFT.

Solution:

Mod-4 reduction of x is given by:

$$x = [3, 3, 3, 3] \quad (5)$$

Now, computing the 4-point DFT of the reduced signal:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (6)$$

This question basically shows that to compute reduced points DFT, taking the mod-N version of the main sequence provides you with a more efficient way of computing the DFT.

3. Compute the 8 point FFT of a length 8 signal which are 8 samples of the signal:

$$x[n] = 4 \cos\left(\frac{\pi n}{2}\right) + \cos(\pi n) \quad (7)$$

Discuss if the FFT accurately represents the entire spectrum of n .

Solution:

The 8 points sequence to be taken is given by:

$$x = [0, 0, 16, 0, 8, 0, 16, 0] \quad (8)$$

Now, computing the 8-point DFT of the signal:

4. Find the DTFT of $y_1[n] = \begin{cases} 1 & \text{if } -N \leq n \leq N \\ 0 & \text{otherwise} \end{cases}$.

Solution:

The DTFT of a signal is given by:

$$Y_1(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y_1[n] e^{-j\omega n} \quad (9)$$

5. Let $X(k)$ be a 14-point DFT of a length 14 real sequence $x(n)$. The first 8 samples of $X(k)$ are given by:

$$X(0) = 12 \quad (10)$$

$$X(1) = -1 + j3 \quad (11)$$

$$X(2) = 3 + j4 \quad (12)$$

$$X(3) = 1 - j5 \quad (13)$$

$$X(4) = -2 + j2 \quad (14)$$

$$X(5) = 6 + j3 \quad (15)$$

$$X(6) = -2 - j3 \quad (16)$$

$$X(7) = 10 \quad (17)$$

- (a) Find the remaining 6 values of $X(k)$.

Solution:

Using the property of DFT for real sequences, we know that:

$$X(N - k) = X^*(k) \quad (18)$$

where N is the length of the sequence.

Thus, we can compute the remaining values as follows:

$$X(8) = X^*(6) = -2 + j3 \quad (19)$$

$$X(9) = X^*(5) = 6 - j3 \quad (20)$$

$$X(10) = X^*(4) = -2 - j2 \quad (21)$$

$$X(11) = X^*(3) = 1 + j5 \quad (22)$$

$$X(12) = X^*(2) = 3 - j4 \quad (23)$$

$$X(13) = X^*(1) = -1 - j3 \quad (24)$$

(b) Find $x(0)$ without computing the IDFT of $X(k)$.

Solution:

Using the property of DFT, we know that:

$$x(0) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) \quad (25)$$

where N is the length of the sequence.

Thus, we can compute $x(0)$ as follows:

(c) Find $x(7)$ without computing the IDFT of $X(k)$.

Solution:

Using the property of DFT, we know that:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} kn} \quad (26)$$

where N is the length of the sequence.

Thus, we can compute $x(7)$ as follows:

6. A length 8 sequence $x(n) = \{-4, 5, 2, -3, 0, -2, 3, 4\}$ has an 8-point DFT $X(k)$. A new sequence is defined as:

$$Y(k) = W_4^{3k} X(k) \quad (27)$$

Find $y(n)$ without computing the IDFT of $Y(k)$.

Solution:

We know that for a DFT, multiplying by a twiddle factor corresponds to a circular shift in the time domain.

Thus, we can compute $y(n)$ as follows:

$$y(n) = x((n - 3) \bmod 8) \quad (28)$$

This process exists because multiplying by $W_N^{kn_0}$ in the frequency domain corresponds to a circular shift by n_0 in the time domain. However, when N is different from the length of the sequence, the circular shift is scaled accordingly. In this case, since we are multiplying by W_4^{3k} instead of W_8^{3k} , the effective shift in the time domain is scaled by a factor of 2 (since $8/4 = 2$). Therefore, the circular shift is by $3*2 = 6$ positions.

7. Let $X(z) = Z\{x(n)\}$, where $x(n) = 0.4^n u(n)$:

- (a) Find $g(n) = Z^{-1}\{X(z^2)\}$ without computing $X(z)$.
 - (b) Find $g(n) = Z^{-1}\{(1 + Z^{-1})X(z^2)\}$ without computing $X(z)$.
8. Let $X(z) = Z\{x(n)\}$, $x(n)$ is a 12 length sequence, and $X_0(k)$ is the 9 point DFT of $x(n)$. Find the inverse DFT of $X_0(k)$ in terms of $x(n)$ without computing $X_0(k)$.
 9. Let $X(e^{j\omega}) = F.T.\{1, 1, \dots, 1\}$ from 0 to 7. $X_1(k)$ is obtained by sampling $X(e^{j\omega})$ at uniform intervals of $\frac{2\pi}{8}$. $X_2(k)$ is obtained by sampling $X(e^{j\omega})$ at uniform intervals of $\frac{2\pi}{5}$.